

CD54/74HC240, CD54/74HCT240, CD74HC241, CD54/74HCT241, CD54/74HC244, CD54/74HCT244

High-Speed CMOS Logic Octal Buffer/Line Drivers, Three-State

Features

- HC/HCT240 Inverting
- HC/HCT241 Non-Inverting
- HC/HCT244 Non-Inverting
- Typical Propagation Delay = 8ns at $V_{CC} = 5V$, $C_L = 15pF$, $T_A = 25^{\circ}C$ for HC240
- Three-State Outputs
- Buffered Inputs
- High-Current Bus Driver Outputs
- Fanout (Over Temperature Range)
 - Standard Outputs 10 LSTTL Loads
 - Bus Driver Outputs 15 LSTTL Loads
- Wide Operating Temperature Range . . . $-55^{\circ}C$ to $125^{\circ}C$
- Balanced Propagation Delay and Transition Times
- Significant Power Reduction Compared to LSTTL Logic ICs
- HC Types
 - 2V to 6V Operation
 - High Noise Immunity: $N_{IL} = 30\%$, $N_{IH} = 30\%$ of V_{CC} at $V_{CC} = 5V$
- HCT Types
 - 4.5V to 5.5V Operation
 - Direct LSTTL Input Logic Compatibility, $V_{IL} = 0.8V$ (Max), $V_{IH} = 2V$ (Min)
 - CMOS Input Compatibility, $I_I \leq 1\mu A$ at V_{OL} , V_{OH}

Description

The 'HC240 and 'HCT240 are inverting three-state buffers having two active-low output enables. The CD74HC241, 'HCT241, 'HC244 and 'HCT244 are non-inverting three-state buffers that differ only in that the 241 has one active-high and one active-low output enable, and the 244 has two active-low output enables. All three types have identical pinouts.

Ordering Information

PART NUMBER	TEMP. RANGE ($^{\circ}C$)	PACKAGE
CD54HC240F3A	-55 to 125	20 Ld CERDIP
CD54HC244F3A	-55 to 125	20 Ld CERDIP
CD54HCT240F3A	-55 to 125	20 Ld CERDIP
CD54HCT241F3A	-55 to 125	20 Ld CERDIP
CD54HCT244F3A	-55 to 125	20 Ld CERDIP
CD74HC240E	-55 to 125	20 Ld PDIP
CD74HC240M	-55 to 125	20 Ld SOIC
CD74HC240M96	-55 to 125	20 Ld SOIC
CD74HC241E	-55 to 125	20 Ld PDIP
CD74HC241M	-55 to 125	20 Ld SOIC
CD74HC241M96	-55 to 125	20 Ld SOIC
CD74HC244E	-55 to 125	20 Ld PDIP
CD74HC244M	-55 to 125	20 Ld SOIC
CD74HC244M96	-55 to 125	20 Ld SOIC
CD74HCT240E	-55 to 125	20 Ld PDIP
CD74HCT240M	-55 to 125	20 Ld SOIC
CD74HCT240M96	-55 to 125	20 Ld SOIC
CD74HCT240PW	-55 to 125	20 Ld TSSOP
CD74HCT240PWR	-55 to 125	20 Ld TSSOP
CD74HCT240PWT	-55 to 125	20 Ld TSSOP
CD74HCT241E	-55 to 125	20 Ld PDIP
CD74HCT241M	-55 to 125	20 Ld SOIC
CD74HCT241M96	-55 to 125	20 Ld SOIC
CD74HCT244E	-55 to 125	20 Ld PDIP
CD74HCT244M	-55 to 125	20 Ld SOIC
CD74HCT244M96	-55 to 125	20 Ld SOIC

NOTE: When ordering, use the entire part number. The suffixes 96 and R denote tape and reel. The suffix T denotes a small-quantity reel of 250.

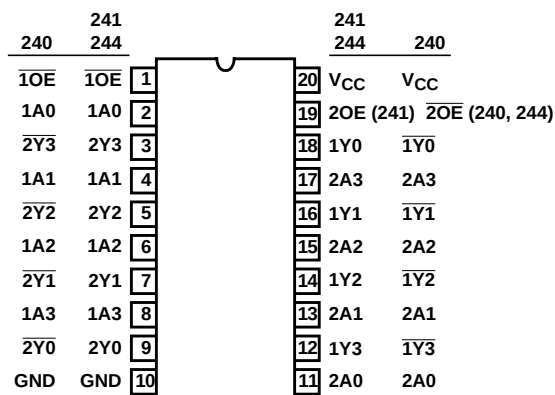
Pinout

CD54HC240, CD54HCT240, CD54HCT241,
CD54HC244, CD54HCT244
(CERDIP)

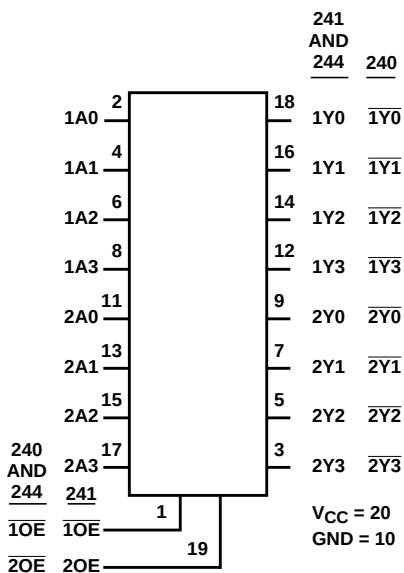
CD74HC240, CD74HC241, CD74HCT241,
CD74HC244, CD74HCT244
(PDIP, SOIC)

CD74HCT240,
(PDIP, SOIC, TSSOP)

TOP VIEW



Functional Diagram



Absolute Maximum Ratings

DC Supply Voltage, V_{CC} -0.5V to 7V
 DC Input Diode Current, I_{IK}
 For $V_I < -0.5V$ or $V_I > V_{CC} + 0.5V$ $\pm 20mA$
 DC Output Diode Current, I_{OK}
 For $V_O < -0.5V$ or $V_O > V_{CC} + 0.5V$ $\pm 20mA$
 DC Drain Current, per Output, I_O
 For $-0.5V < V_O < V_{CC} + 0.5V$ $\pm 35mA$
 DC Output Source or Sink Current per Output Pin, I_O
 For $V_O > -0.5V$ or $V_O < V_{CC} + 0.5V$ $\pm 25mA$
 DC V_{CC} or Ground Current, I_{CC} $\pm 70mA$

Thermal Information

Thermal Resistance (Typical, Note 1) θ_{JA}
 E (PDIP) Package $69^{\circ}C/W$
 M (SOIC) Package $58^{\circ}C/W$
 PW (TSSOP) Package $83^{\circ}C/W$
 Maximum Junction Temperature $150^{\circ}C$
 Maximum Storage Temperature Range $-65^{\circ}C$ to $150^{\circ}C$
 Maximum Lead Temperature (Soldering 10s) $300^{\circ}C$
 (SOIC - Lead Tips Only)

Operating Conditions

Temperature Range (T_A) $-55^{\circ}C$ to $125^{\circ}C$
 Supply Voltage Range, V_{CC}
 HC Types 2V to 6V
 HCT Types 4.5V to 5.5V
 DC Input or Output Voltage, V_I , V_O 0V to V_{CC}
 Input Rise and Fall Time
 2V 1000ns (Max)
 4.5V 500ns (Max)
 6V 400ns (Max)

CAUTION: Stresses above those listed in "Absolute Maximum Ratings" may cause permanent damage to the device. This is a stress only rating and operation of the device at these or any other conditions above those indicated in the operational sections of this specification is not implied.

NOTE:

1. The package thermal impedance is calculated in accordance with JESD 51-7.

DC Electrical Specifications

PARAMETER	SYMBOL	TEST CONDITIONS		V _{CC} (V)	25°C			-40°C TO 85°C		-55°C TO 125°C		UNITS
		V _I (V)	I _O (mA)		MIN	TYP	MAX	MIN	MAX	MIN	MAX	
HC TYPES												
High Level Input Voltage	V _{IH}	-	-	2	1.5	-	-	1.5	-	1.5	-	V
				4.5	3.15	-	-	3.15	-	3.15	-	V
				6	4.2	-	-	4.2	-	4.2	-	V
Low Level Input Voltage	V _{IL}	-	-	2	-	-	0.5	-	0.5	-	0.5	V
				4.5	-	-	1.35	-	1.35	-	1.35	V
				6	-	-	1.8	-	1.8	-	1.8	V
High Level Output Voltage CMOS Loads	V _{OH}	V _{IH} or V _{IL}	-0.02	2	1.9	-	-	1.9	-	1.9	-	V
			-0.02	4.5	4.4	-	-	4.4	-	4.4	-	V
			-0.02	6	5.9	-	-	5.9	-	5.9	-	V
High Level Output Voltage TTL Loads			-6	4.5	3.98	-	-	3.84	-	3.7	-	V
			-7.8	6	5.48	-	-	5.34	-	5.2	-	V
Low Level Output Voltage CMOS Loads	V _{OL}	V _{IH} or V _{IL}	0.02	2	-	-	0.1	-	0.1	-	0.1	V
			0.02	4.5	-	-	0.1	-	0.1	-	0.1	V
			0.02	6	-	-	0.1	-	0.1	-	0.1	V
Low Level Output Voltage TTL Loads			6	4.5	-	-	0.26	-	0.33	-	0.4	V
			7.8	6	-	-	0.26	-	0.33	-	0.4	V
Input Leakage Current	I _I	V _{CC} or GND	-	6	-	-	±0.1	-	±1	-	±1	μA
Quiescent Device Current	I _{CC}	V _{CC} or GND	0	6	-	-	8	-	80	-	160	μA

DC Electrical Specifications (Continued)

PARAMETER	SYMBOL	TEST CONDITIONS		V_{CC} (V)	25°C			-40°C TO 85°C		-55°C TO 125°C		UNITS
		V_I (V)	I_O (mA)		MIN	TYP	MAX	MIN	MAX	MIN	MAX	
Three-State Leakage Current	I_{OZ}	V_{IL} or V_{IH}	-	6	-	-	± 0.5	-	± 0.5	-	± 10	μA
HCT TYPES												
High Level Input Voltage	V_{IH}	-	-	4.5 to 5.5	2	-	-	2	-	2	-	V
Low Level Input Voltage	V_{IL}	-	-	4.5 to 5.5	-	-	0.8	-	0.8	-	0.8	V
High Level Output Voltage CMOS Loads	V_{OH}	V_{IH} or V_{IL}	-0.02	4.5	4.4	-	-	4.4	-	4.4	-	V
High Level Output Voltage TTL Loads			-6	4.5	3.98	-	-	3.84	-	3.7	-	V
Low Level Output Voltage CMOS Loads	V_{OL}	V_{IH} or V_{IL}	0.02	4.5	-	-	0.1	-	0.1	-	0.1	V
Low Level Output Voltage TTL Loads			6	4.5	-	-	0.26	-	0.33	-	0.4	V
Input Leakage Current	I_I	V_{CC} to GND	0	5.5	-	-	± 0.1	-	± 1	-	± 1	μA
Quiescent Device Current	I_{CC}	V_{CC} or GND	0	5.5	-	-	8	-	80	-	160	μA
Additional Quiescent Device Current Per Input Pin: 1 Unit Load	ΔI_{CC} (Note 2)	V_{CC} -2.1	-	4.5 to 5.5	-	100	360	-	450	-	490	μA
Three-State Leakage Current	I_{OZ}	V_{IL} or V_{IH}	-	5.5	-	-	± 0.5	-	± 5	-	± 10	μA

NOTE:

2. For dual-supply systems theoretical worst case ($V_I = 2.4V$, $V_{CC} = 5.5V$) specification is 1.8mA.

HCT Input Loading Table

INPUT	UNIT LOADS
HCT240	
nA0-A3	1.5
$\overline{1OE}$	0.7
$\overline{2OE}$	0.7
HCT241	
nA0-A3	0.7
$\overline{1OE}$	0.7
$\overline{2OE}$	1.5
HCT244	
nA0-A3	0.7
$\overline{1OE}$	0.7
$\overline{2OE}$	0.7

NOTE: Unit Load is ΔI_{CC} limit specified in DC Electrical Specifications table, e.g., 360 μA max at 25°C.

Switching Specifications $C_L = 50\text{pF}$, Input $t_r, t_f = 6\text{ns}$

PARAMETER	SYMBOL	TEST CONDI- TIONS	V _{CC} (V)	25°C			-40°C TO 85°C			-55°C TO 125°C			UNITS
				MIN	TYP	MAX	MIN	TYP	MAX	MIN	TYP	MAX	
HC TYPES													
Propagation Delay Data to Outputs HC240	t _{PLH} , t _{PHL}	C _L = 50pF	2	-	-	100	-	-	125	-	-	150	ns
			4.5	-	-	20	-	-	25	-	-	30	ns
		C _L = 15pF	5	-	8	-	-	-	-	-	-	-	ns
		C _L = 50pF	6	-	-	17	-	-	21	-	-	26	ns
Data to Outputs HC241	t _{PLH} , t _{PHL}	C _L = 50pF	2	-	-	110	-	-	140	-	-	165	ns
			4.5	-	-	22	-	-	28	-	-	33	ns
		C _L = 15pF	5	-	9	-	-	-	-	-	-	-	ns
		C _L = 50pF	6	-	-	19	-	-	24	-	-	28	ns
Data to Outputs HC244	t _{PLH} , t _{PHL}	C _L = 50pF	2	-	-	110	-	-	140	-	-	165	ns
			4.5	-	-	22	-	-	28	-	-	33	ns
		C _L = 15pF	5	-	9	-	-	-	-	-	-	-	ns
		C _L = 50pF	6	-	-	19	-	-	24	-	-	28	ns
Output Enable and Disable Time	t _{THL} , t _{TLH}	C _L = 50pF	2	-	-	150	-	-	190	-	-	225	ns
			4.5	-	-	30	-	-	38	-	-	45	ns
			5	-	12	-	-	-	-	-	-	-	ns
			6	-	-	26	-	-	33	-	-	38	ns
Output Transition Time	t _{TLH} , t _{THL}	C _L = 50pF	2	-	-	60	-	-	75	-	-	90	ns
			4.5	-	-	12	-	-	15	-	-	18	ns
			6	-	-	10	-	-	13	-	-	15	ns
Input Capacitance	C _I	C _L = 50pF	-	10	-	10	-	-	10	-	-	10	pF
Three-State Output Capacitance	C _O	C _L = 50pF	-	-	-	20	-	-	20	-	-	20	pF
Power Dissipation Capacitance (Notes 3, 4)	C _{PD}	C _L = 15pF											
HC240			5	-	38	-	-	-	-	-	-	-	pF
HC241			5	-	34	-	-	-	-	-	-	-	pF
HC244			5	-	46	-	-	-	-	-	-	-	pF
HCT TYPES													
Propagation Delay Data to Outputs HCT240	t _{PHL} , t _{PLH}	C _L = 50pF	4.5	-	-	22	-	-	28	-	-	33	ns
		C _L = 15pF	5	-	9	-	-	-	-	-	-	-	ns
Data to Outputs HCT241	t _{PHL} , t _{PLH}	C _L = 50pF	4.5	-	-	25	-	-	31	-	-	38	ns
		C _L = 15pF	5	-	10	-	-	-	-	-	-	-	ns
Data to Outputs HCT244	t _{PHL} , t _{PLH}	C _L = 50pF	4.5	-	-	25	-	-	31	-	-	38	ns
		C _L = 15pF	5	-	10	-	-	-	-	-	-	-	ns

Switching Specifications $C_L = 50\text{pF}$, Input $t_r, t_f = 6\text{ns}$ (Continued)

PARAMETER	SYMBOL	TEST CONDITIONS	V_{CC} (V)	25°C			-40°C TO 85°C			-55°C TO 125°C			UNITS
				MIN	TYP	MAX	MIN	TYP	MAX	MIN	TYP	MAX	
Output Enable and Disable Times	t_{TLH}, t_{THL}	$C_L = 50\text{pF}$	4.5	-	-	30	-	-	38	-	-	45	ns
Output Transition Time	t_{THL}, t_{TLH}	$C_L = 50\text{pF}$	4.5	-	-	12	-	-	15	-	-	18	ns
Input Capacitance	C_I	$C_L = 50\text{pF}$	-	10	-	10	-	-	10	-	-	10	pF
Power Dissipation Capacitance (Notes 3, 4)	C_{PD}												
HCT240		-	5	-	40	-	-	-	-	-	-	-	pF
HCT241		-	5	-	38	-	-	-	-	-	-	-	pF
HCT244		-	5	-	40	-	-	-	-	-	-	-	pF

NOTES:

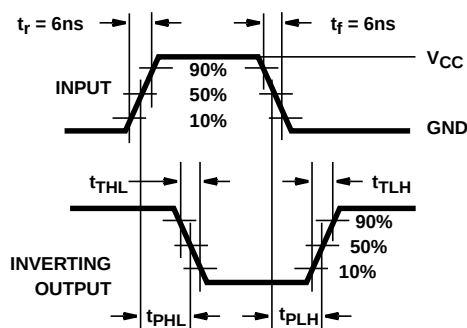
3. C_{PD} is used to determine the dynamic power consumption, per channel.4. $P_D = V_{CC}^2 f_i (C_{PD} + C_L)$ where f_i = Input Frequency, f_o = Output Frequency, C_L = Output Load Capacitance, V_{CC} = Supply Voltage.**Test Circuits and Waveforms**

FIGURE 1. HC TRANSITION TIMES AND PROPAGATION DELAY TIMES, COMBINATION LOGIC

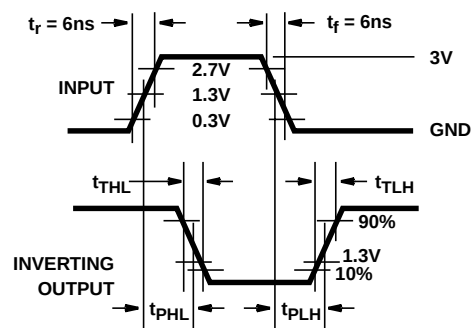


FIGURE 2. HCT TRANSITION TIMES AND PROPAGATION DELAY TIMES, COMBINATION LOGIC

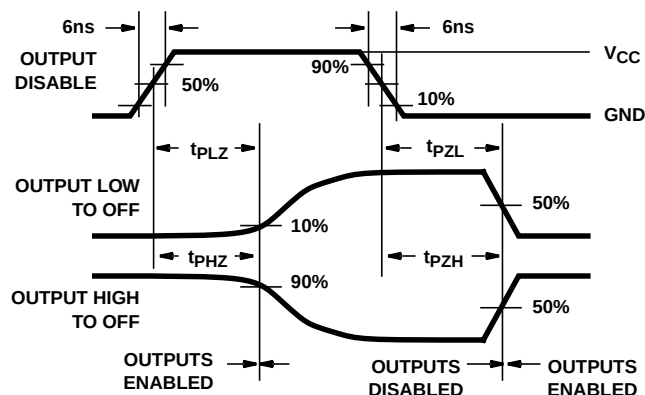


FIGURE 3. HC THREE-STATE PROPAGATION DELAY WAVEFORM

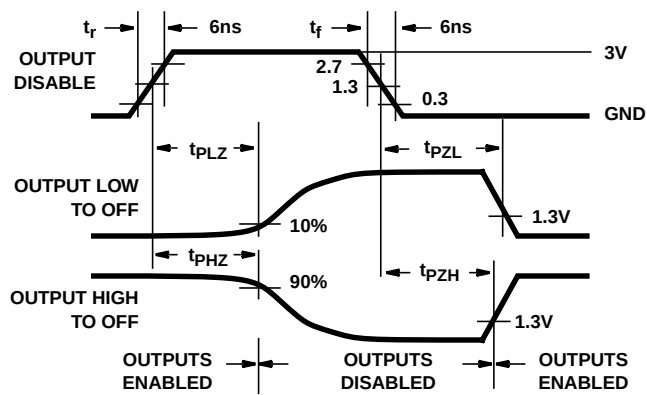
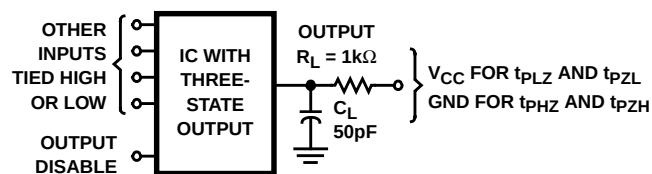


FIGURE 4. HCT THREE-STATE PROPAGATION DELAY WAVEFORM

Test Circuits and Waveforms (Continued)



NOTE: Open drain waveforms t_{PLZ} and t_{PZL} are the same as those for three-state shown on the left. The test circuit is Output $R_L = 1k\Omega$ to V_{CC} , $C_L = 50pF$.

FIGURE 5. HC AND HCT THREE-STATE PROPAGATION DELAY TEST CIRCUIT

PACKAGING INFORMATION

Orderable Device	Status ⁽¹⁾	Package Type	Package Drawing	Pins	Package Qty	Eco Plan ⁽²⁾	Lead/Ball Finish	MSL Peak Temp ⁽³⁾
CD54HC240F3A	ACTIVE	CDIP	J	20	1	TBD	Call TI	Level-NC-NC-NC
CD54HC244F	ACTIVE	CDIP	J	20	1	TBD	Call TI	Level-NC-NC-NC
CD54HC244F3A	ACTIVE	CDIP	J	20	1	TBD	Call TI	Level-NC-NC-NC
CD54HCT240F3A	ACTIVE	CDIP	J	20	1	TBD	Call TI	Level-NC-NC-NC
CD54HCT241F3A	ACTIVE	CDIP	J	20	1	TBD	Call TI	Level-NC-NC-NC
CD54HCT244F	ACTIVE	CDIP	J	20	1	TBD	Call TI	Level-NC-NC-NC
CD54HCT244F3A	ACTIVE	CDIP	J	20	1	TBD	Call TI	Level-NC-NC-NC
CD74HC240E	ACTIVE	PDIP	N	20	20	Pb-Free (RoHS)	CU NIPDAU	Level-NC-NC-NC
CD74HC240EE4	ACTIVE	PDIP	N	20	20	Pb-Free (RoHS)	CU NIPDAU	Level-NC-NC-NC
CD74HC240M	ACTIVE	SOIC	DW	20	25	Pb-Free (RoHS)	CU NIPDAU	Level-2-250C-1 YEAR/ Level-1-235C-UNLIM
CD74HC240M96	ACTIVE	SOIC	DW	20	2000	Pb-Free (RoHS)	CU NIPDAU	Level-2-250C-1 YEAR/ Level-1-235C-UNLIM
CD74HC240M96E4	ACTIVE	SOIC	DW	20	2000	Pb-Free (RoHS)	CU NIPDAU	Level-2-250C-1 YEAR/ Level-1-235C-UNLIM
CD74HC240ME4	ACTIVE	SOIC	DW	20	25	Pb-Free (RoHS)	CU NIPDAU	Level-2-250C-1 YEAR/ Level-1-235C-UNLIM
CD74HC241E	ACTIVE	PDIP	N	20	20	Pb-Free (RoHS)	CU NIPDAU	Level-NC-NC-NC
CD74HC241EE4	ACTIVE	PDIP	N	20	20	Pb-Free (RoHS)	CU NIPDAU	Level-NC-NC-NC
CD74HC241M	ACTIVE	SOIC	DW	20	25	Pb-Free (RoHS)	CU NIPDAU	Level-2-250C-1 YEAR/ Level-1-235C-UNLIM
CD74HC241M96	ACTIVE	SOIC	DW	20	2000	Pb-Free (RoHS)	CU NIPDAU	Level-2-250C-1 YEAR/ Level-1-235C-UNLIM
CD74HC241M96E4	ACTIVE	SOIC	DW	20	2000	Pb-Free (RoHS)	CU NIPDAU	Level-2-250C-1 YEAR/ Level-1-235C-UNLIM
CD74HC241ME4	ACTIVE	SOIC	DW	20	25	Pb-Free (RoHS)	CU NIPDAU	Level-2-250C-1 YEAR/ Level-1-235C-UNLIM
CD74HC244E	ACTIVE	PDIP	N	20	20	Pb-Free (RoHS)	CU NIPDAU	Level-NC-NC-NC
CD74HC244EE4	ACTIVE	PDIP	N	20	20	Pb-Free (RoHS)	CU NIPDAU	Level-NC-NC-NC
CD74HC244M	ACTIVE	SOIC	DW	20	25	Pb-Free (RoHS)	CU NIPDAU	Level-2-250C-1 YEAR/ Level-1-235C-UNLIM
CD74HC244M96	ACTIVE	SOIC	DW	20	2000	Pb-Free (RoHS)	CU NIPDAU	Level-2-250C-1 YEAR/ Level-1-235C-UNLIM
CD74HC244M96E4	ACTIVE	SOIC	DW	20	2000	Pb-Free (RoHS)	CU NIPDAU	Level-2-250C-1 YEAR/ Level-1-235C-UNLIM
CD74HC244ME4	ACTIVE	SOIC	DW	20	25	Pb-Free (RoHS)	CU NIPDAU	Level-2-250C-1 YEAR/ Level-1-235C-UNLIM
CD74HCT240E	ACTIVE	PDIP	N	20	20	Pb-Free (RoHS)	CU NIPDAU	Level-NC-NC-NC
CD74HCT240EE4	ACTIVE	PDIP	N	20	20	Pb-Free (RoHS)	CU NIPDAU	Level-NC-NC-NC
CD74HCT240M	ACTIVE	SOIC	DW	20	25	Pb-Free	CU NIPDAU	Level-2-250C-1 YEAR/

Orderable Device	Status ⁽¹⁾	Package Type	Package Drawing	Pins	Package Qty	Eco Plan ⁽²⁾	Lead/Ball Finish	MSL Peak Temp ⁽³⁾
						(RoHS)		Level-1-235C-UNLIM
CD74HCT240M96	ACTIVE	SOIC	DW	20	2000	Pb-Free (RoHS)	CU NIPDAU	Level-2-250C-1 YEAR/ Level-1-235C-UNLIM
CD74HCT240M96E4	ACTIVE	SOIC	DW	20	2000	Pb-Free (RoHS)	CU NIPDAU	Level-2-250C-1 YEAR/ Level-1-235C-UNLIM
CD74HCT240ME4	ACTIVE	SOIC	DW	20	25	Pb-Free (RoHS)	CU NIPDAU	Level-2-250C-1 YEAR/ Level-1-235C-UNLIM
CD74HCT240PW	ACTIVE	TSSOP	PW	20	70	Pb-Free (RoHS)	CU NIPDAU	Level-1-250C-UNLIM
CD74HCT240PWE4	ACTIVE	TSSOP	PW	20	70	Pb-Free (RoHS)	CU NIPDAU	Level-1-250C-UNLIM
CD74HCT240PWR	ACTIVE	TSSOP	PW	20	2000	Pb-Free (RoHS)	CU NIPDAU	Level-1-250C-UNLIM
CD74HCT240PWRE4	ACTIVE	TSSOP	PW	20	2000	Pb-Free (RoHS)	CU NIPDAU	Level-1-250C-UNLIM
CD74HCT240PWT	ACTIVE	TSSOP	PW	20	250	Pb-Free (RoHS)	CU NIPDAU	Level-1-250C-UNLIM
CD74HCT240PWTE4	ACTIVE	TSSOP	PW	20	250	Pb-Free (RoHS)	CU NIPDAU	Level-1-250C-UNLIM
CD74HCT241E	ACTIVE	PDIP	N	20	20	Pb-Free (RoHS)	CU NIPDAU	Level-NC-NC-NC
CD74HCT241M	ACTIVE	SOIC	DW	20	25	Pb-Free (RoHS)	CU NIPDAU	Level-2-250C-1 YEAR/ Level-1-235C-UNLIM
CD74HCT241M96	ACTIVE	SOIC	DW	20	2000	Pb-Free (RoHS)	CU NIPDAU	Level-2-250C-1 YEAR/ Level-1-235C-UNLIM
CD74HCT241M96E4	ACTIVE	SOIC	DW	20	2000	Pb-Free (RoHS)	CU NIPDAU	Level-2-250C-1 YEAR/ Level-1-235C-UNLIM
CD74HCT241ME4	ACTIVE	SOIC	DW	20	25	Pb-Free (RoHS)	CU NIPDAU	Level-2-250C-1 YEAR/ Level-1-235C-UNLIM
CD74HCT244E	ACTIVE	PDIP	N	20	20	Pb-Free (RoHS)	CU NIPDAU	Level-NC-NC-NC
CD74HCT244EE4	ACTIVE	PDIP	N	20	20	Pb-Free (RoHS)	CU NIPDAU	Level-NC-NC-NC
CD74HCT244M	ACTIVE	SOIC	DW	20	25	Pb-Free (RoHS)	CU NIPDAU	Level-2-250C-1 YEAR/ Level-1-235C-UNLIM
CD74HCT244M96	ACTIVE	SOIC	DW	20	2000	Pb-Free (RoHS)	CU NIPDAU	Level-2-250C-1 YEAR/ Level-1-235C-UNLIM
CD74HCT244M96E4	ACTIVE	SOIC	DW	20	2000	Pb-Free (RoHS)	CU NIPDAU	Level-2-250C-1 YEAR/ Level-1-235C-UNLIM
CD74HCT244ME4	ACTIVE	SOIC	DW	20	25	Pb-Free (RoHS)	CU NIPDAU	Level-2-250C-1 YEAR/ Level-1-235C-UNLIM

⁽¹⁾ The marketing status values are defined as follows:

ACTIVE: Product device recommended for new designs.

LIFEBUY: TI has announced that the device will be discontinued, and a lifetime-buy period is in effect.

NRND: Not recommended for new designs. Device is in production to support existing customers, but TI does not recommend using this part in a new design.

PREVIEW: Device has been announced but is not in production. Samples may or may not be available.

OBSOLETE: TI has discontinued the production of the device.

⁽²⁾ Eco Plan - The planned eco-friendly classification: Pb-Free (RoHS) or Green (RoHS & no Sb/Br) - please check <http://www.ti.com/productcontent> for the latest availability information and additional product content details.

TBD: The Pb-Free/Green conversion plan has not been defined.

Pb-Free (RoHS): TI's terms "Lead-Free" or "Pb-Free" mean semiconductor products that are compatible with the current RoHS requirements for all 6 substances, including the requirement that lead not exceed 0.1% by weight in homogeneous materials. Where designed to be soldered at high temperatures, TI Pb-Free products are suitable for use in specified lead-free processes.

Green (RoHS & no Sb/Br): TI defines "Green" to mean Pb-Free (RoHS compatible), and free of Bromine (Br) and Antimony (Sb) based flame retardants (Br or Sb do not exceed 0.1% by weight in homogeneous material)

(3) MSL, Peak Temp. -- The Moisture Sensitivity Level rating according to the JEDEC industry standard classifications, and peak solder temperature.

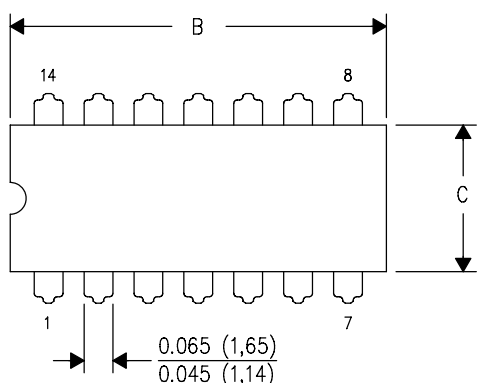
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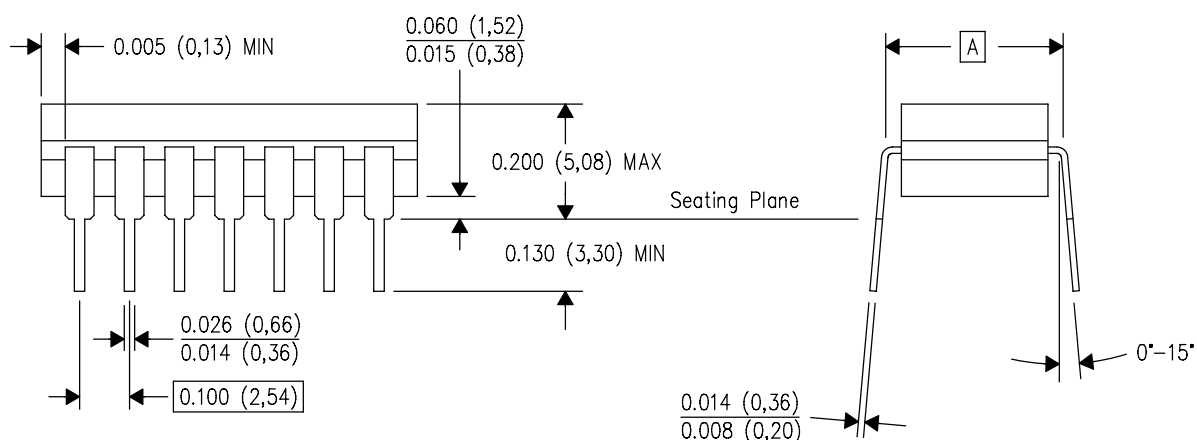
J (R-GDIP-T**)

14 LEADS SHOWN

CERAMIC DUAL IN-LINE PACKAGE



PINS ** DIM	14	16	18	20
A	0.300 (7,62) BSC	0.300 (7,62) BSC	0.300 (7,62) BSC	0.300 (7,62) BSC
B MAX	0.785 (19,94)	.840 (21,34)	0.960 (24,38)	1.060 (26,92)
B MIN	—	—	—	—
C MAX	0.300 (7,62)	0.300 (7,62)	0.310 (7,87)	0.300 (7,62)
C MIN	0.245 (6,22)	0.245 (6,22)	0.220 (5,59)	0.245 (6,22)



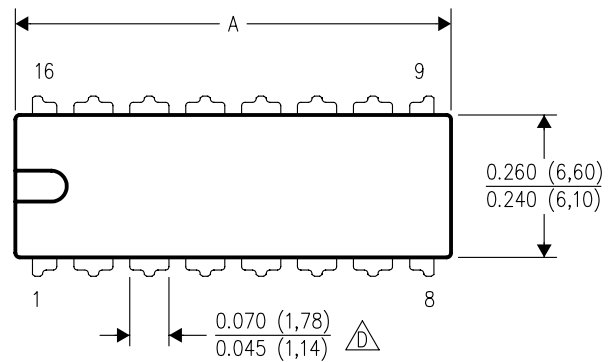
4040083/F 03/03

- NOTES:
- A. All linear dimensions are in inches (millimeters).
 - B. This drawing is subject to change without notice.
 - C. This package is hermetically sealed with a ceramic lid using glass frit.
 - D. Index point is provided on cap for terminal identification only on press ceramic glass frit seal only.
 - E. Falls within MIL STD 1835 GDIP1-T14, GDIP1-T16, GDIP1-T18 and GDIP1-T20.

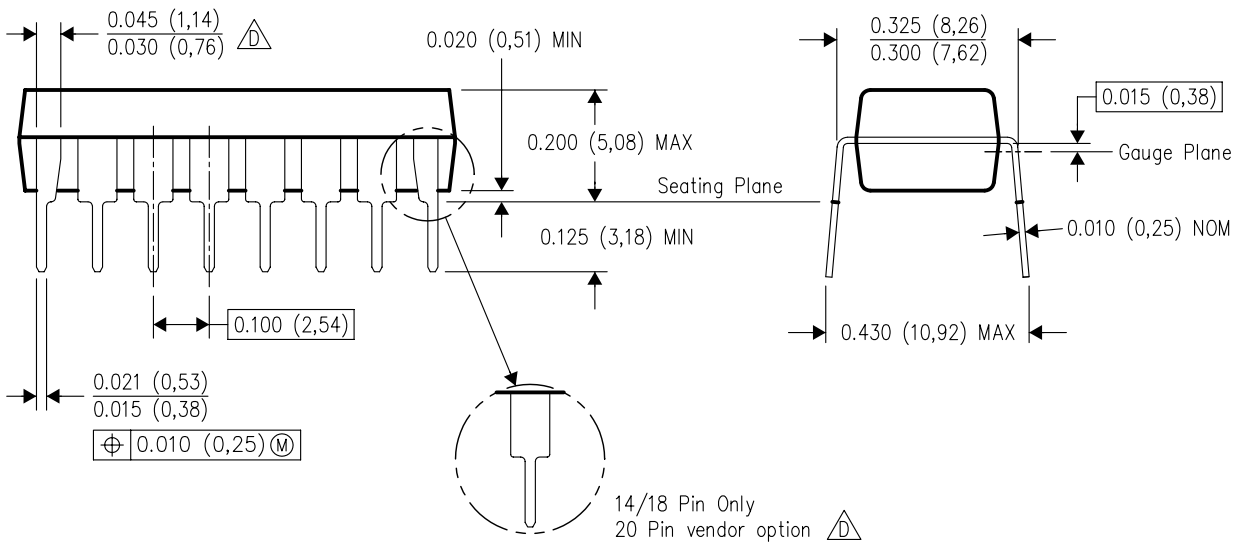
N (R-PDIP-T**)

16 PINS SHOWN

PLASTIC DUAL-IN-LINE PACKAGE



PINS **	14	16	18	20
DIM				
A MAX	0.775 (19,69)	0.775 (19,69)	0.920 (23,37)	1.060 (26,92)
A MIN	0.745 (18,92)	0.745 (18,92)	0.850 (21,59)	0.940 (23,88)
MS-001 VARIATION	AA	BB	AC	AD



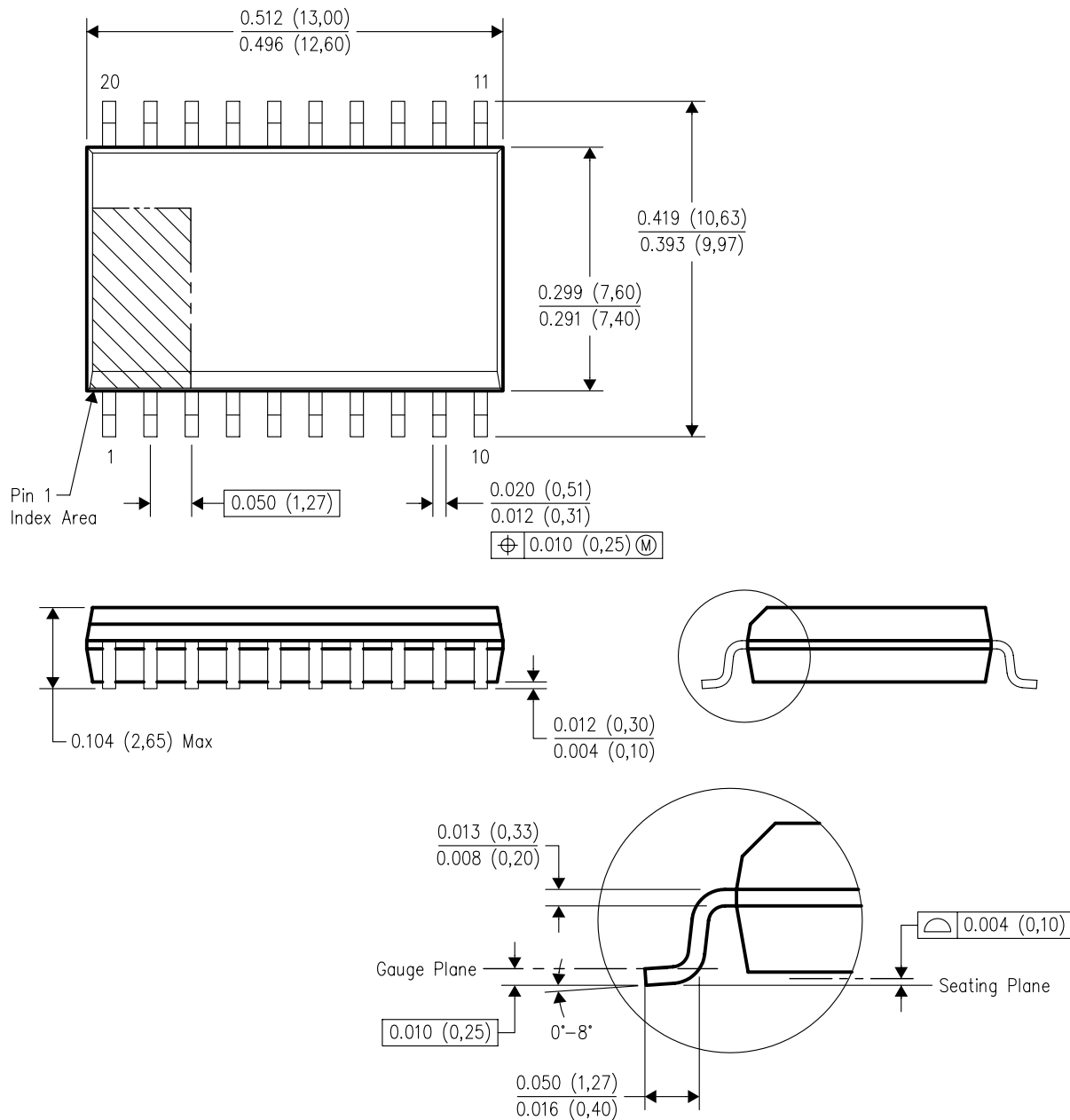
14/18 Pin Only
20 Pin vendor option

4040049/E 12/2002

- NOTES:
- A. All linear dimensions are in inches (millimeters).
 - B. This drawing is subject to change without notice.
 - Falls within JEDEC MS-001, except 18 and 20 pin minimum body length (Dim A).
 - The 20 pin end lead shoulder width is a vendor option, either half or full width.

DW (R-PDSO-G20)

PLASTIC SMALL-OUTLINE PACKAGE



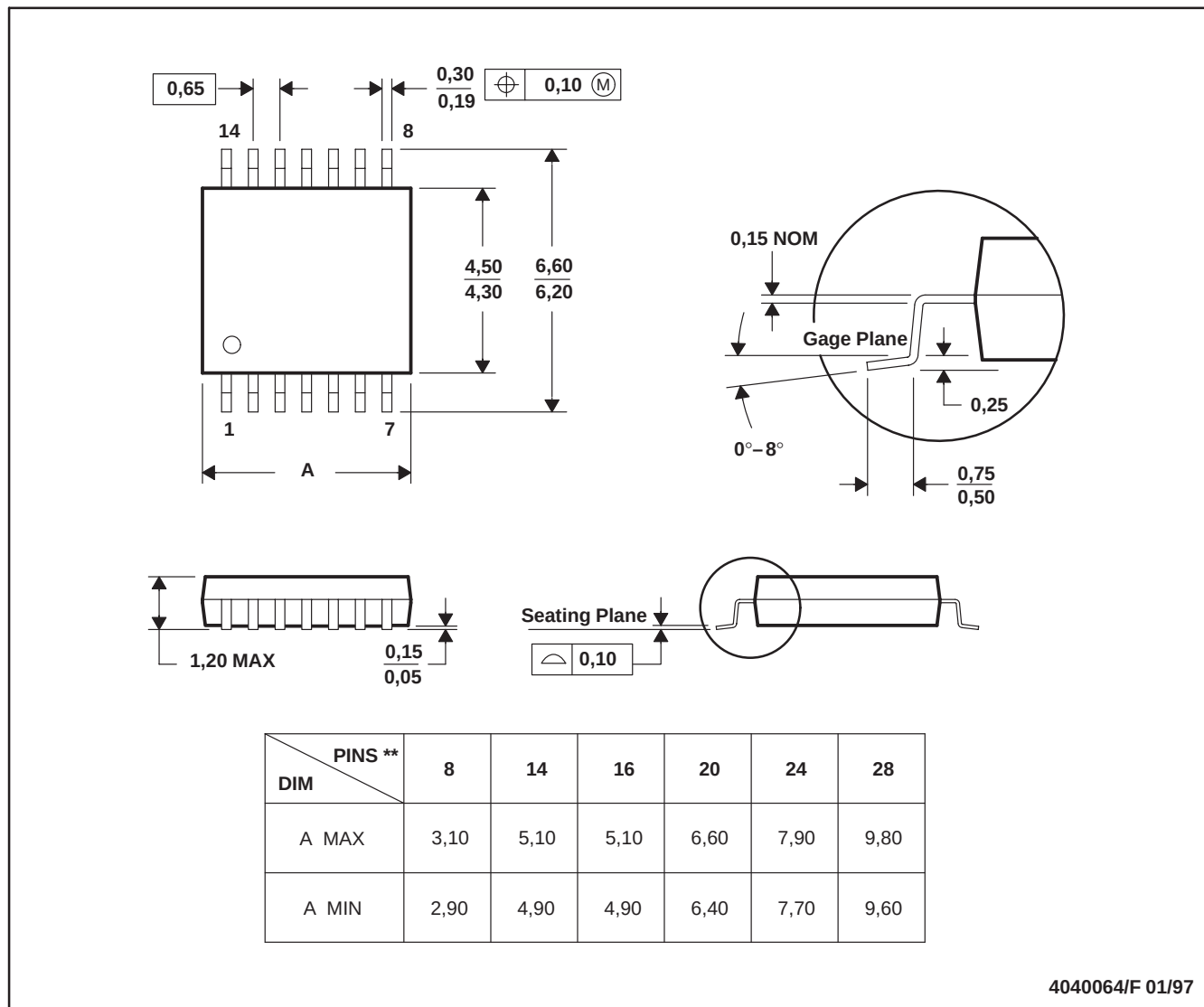
4040000-4/F 06/2004

- NOTES:
- A. All linear dimensions are in inches (millimeters).
 - B. This drawing is subject to change without notice.
 - C. Body dimensions do not include mold flash or protrusion not to exceed 0.006 (0,15).
 - D. Falls within JEDEC MS-013 variation AC.

PW (R-PDSO-G**)

PLASTIC SMALL-OUTLINE PACKAGE

14 PINS SHOWN



- NOTES: A. All linear dimensions are in millimeters.
 B. This drawing is subject to change without notice.
 C. Body dimensions do not include mold flash or protrusion not to exceed 0,15.
 D. Falls within JEDEC MO-153

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